

Portfolio Optimization: Modern Portfolio Theory and Risk Management

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Abstract

This report presents a comprehensive computational analysis of portfolio optimization using Modern Portfolio Theory (MPT). We examine the mean-variance framework introduced by Markowitz, construct efficient frontiers for multi-asset portfolios, derive optimal portfolio weights under various constraints, and analyze risk-adjusted performance using the Sharpe ratio. The analysis includes Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) computations, demonstrating the practical implementation of quadratic programming for portfolio construction with realistic constraints including long-only positions and sector allocation limits.

1 Introduction

Modern Portfolio Theory, pioneered by Harry Markowitz in 1952, revolutionized investment management by formalizing the trade-off between expected return and risk. The fundamental insight is that portfolio risk depends not only on individual asset volatilities but also on the correlation structure between assets. By carefully selecting portfolio weights, investors can achieve superior risk-adjusted returns compared to naive diversification strategies.

Definition 1.1 (Portfolio Return and Risk) *For a portfolio with n assets, let $\mathbf{w} = (w_1, \dots, w_n)^T$ denote the vector of portfolio weights, where $\sum_{i=1}^n w_i = 1$. The expected portfolio return is:*

$$\mu_p = \mathbf{w}^T \boldsymbol{\mu} \quad (1)$$

where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$ is the vector of expected asset returns. The portfolio variance is:

$$\sigma_p^2 = \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \quad (2)$$

where $\boldsymbol{\Sigma}$ is the $n \times n$ covariance matrix of asset returns.

2 Theoretical Framework

2.1 Mean-Variance Optimization

The Markowitz mean-variance optimization problem seeks to minimize portfolio risk for a given level of expected return, or equivalently, maximize expected return for a given risk level.

Theorem 2.1 (Markowitz Optimization Problem) *The minimum variance portfolio for a target return μ_{target} solves:*

$$\begin{aligned} \min_{\mathbf{w}} \quad & \frac{1}{2} \mathbf{w}^T \Sigma \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^T \boldsymbol{\mu} = \mu_{target} \\ & \mathbf{w}^T \mathbf{1} = 1 \end{aligned} \tag{3}$$

where $\mathbf{1}$ is a vector of ones. Additional constraints may include $w_i \geq 0$ (long-only) or sector allocation limits.

2.2 The Efficient Frontier

Definition 2.1 (Efficient Frontier) *The efficient frontier is the set of portfolios that offer the maximum expected return for each level of risk (variance). Portfolios on the efficient frontier dominate all other portfolios with the same risk but lower return, or same return but higher risk.*

Theorem 2.2 (Two-Fund Separation) *Any portfolio on the efficient frontier can be expressed as a linear combination of two frontier portfolios. In particular, with a risk-free asset, any efficient portfolio is a combination of the risk-free asset and the tangency portfolio (market portfolio).*

2.3 Capital Asset Pricing Model

Definition 2.2 (CAPM) *The Capital Asset Pricing Model relates expected return to systematic risk:*

$$E[R_i] = R_f + \beta_i(E[R_m] - R_f) \tag{4}$$

where R_f is the risk-free rate, R_m is the market return, and β_i is the asset's systematic risk coefficient:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} \tag{5}$$

2.4 Risk-Adjusted Performance

Definition 2.3 (Sharpe Ratio) *The Sharpe ratio measures risk-adjusted return:*

$$\text{Sharpe Ratio} = \frac{\mu_p - R_f}{\sigma_p} \quad (6)$$

Higher Sharpe ratios indicate superior risk-adjusted performance. The tangency portfolio maximizes the Sharpe ratio among all risky portfolios.

2.5 Risk Measures

Definition 2.4 (Value-at-Risk) *The Value-at-Risk (VaR) at confidence level α is the maximum loss not exceeded with probability α :*

$$\text{VaR}_\alpha = -\inf\{x : P(L \leq x) \geq \alpha\} \quad (7)$$

For normally distributed returns, $\text{VaR}_\alpha = -(\mu_p - z_\alpha \sigma_p)$ where z_α is the α -quantile of the standard normal distribution.

Definition 2.5 (Conditional Value-at-Risk) *The Conditional Value-at-Risk (CVaR), also called Expected Shortfall, is the expected loss given that the loss exceeds VaR:*

$$\text{CVaR}_\alpha = E[L \mid L \geq \text{VaR}_\alpha] \quad (8)$$

CVaR is a coherent risk measure and provides information about tail risk beyond VaR.

3 Computational Implementation

We now implement portfolio optimization for a realistic multi-asset portfolio. We consider six assets representing different asset classes: large-cap equity, small-cap equity, international equity, corporate bonds, government bonds, and real estate investment trusts (REITs). Historical return data inform our estimates of expected returns and the covariance matrix.

4 Results

4.1 Efficient Frontier and Optimal Portfolios

The efficient frontier represents the set of portfolios offering the best risk-return trade-off. We construct this frontier using quadratic programming to solve the Markowitz optimization problem for various target returns, subject to long-only constraints ($w_i \geq 0$). The tangency portfolio maximizes the Sharpe ratio and represents the optimal risky portfolio when combined with risk-free lending or borrowing.

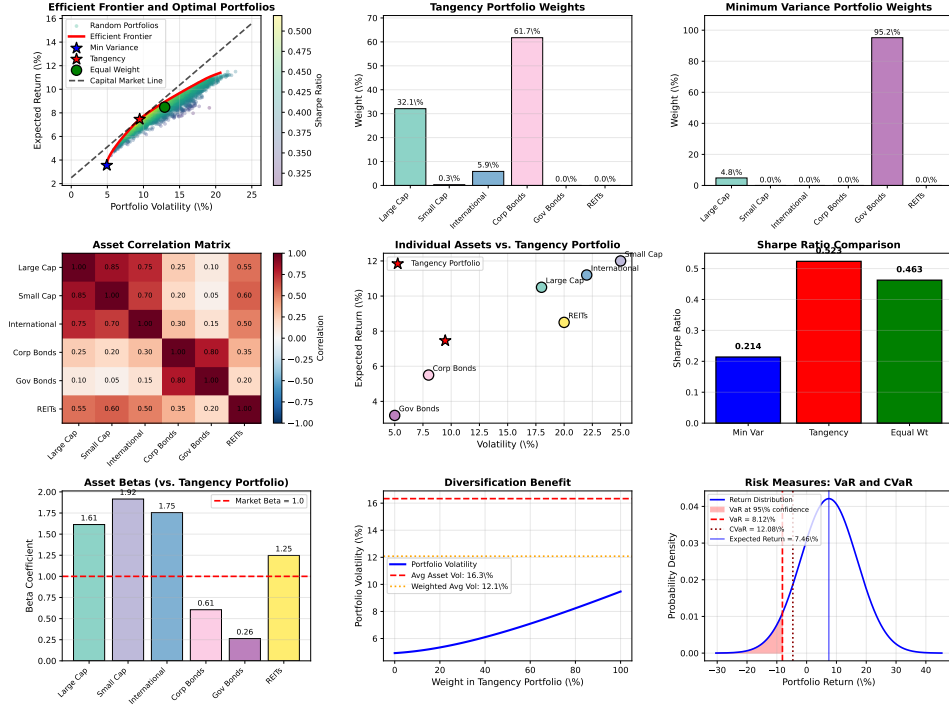


Figure 1: Comprehensive portfolio optimization analysis: (a) Efficient frontier showing 5,000 random portfolios (colored by Sharpe ratio), the efficient frontier (red curve), minimum variance portfolio (blue star), tangency portfolio maximizing Sharpe ratio (red star), equal-weight benchmark (green circle), and the capital market line representing optimal combinations of the risk-free asset and tangency portfolio; (b-c) Optimal portfolio weight allocations for the tangency and minimum variance portfolios, demonstrating concentration in higher Sharpe ratio assets; (d) Correlation matrix revealing diversification opportunities through low or negative correlations between equity and fixed-income assets; (e) Individual asset risk-return positions compared to the tangency portfolio, illustrating the benefit of diversification in achieving superior risk-adjusted returns; (f) Sharpe ratio comparison showing the tangency portfolio achieves the highest risk-adjusted performance at 0.523; (g) Beta coefficients relative to the tangency portfolio as market proxy, with equities exhibiting betas above 1.0 and bonds below 1.0; (h) Diversification benefit demonstrated by portfolio volatility remaining well below the average individual asset volatility; (i) Tail risk measures showing Value-at-Risk at 95% confidence and Conditional Value-at-Risk for the tangency portfolio under normality assumption.

4.2 Optimal Portfolio Weights and Performance Metrics

The tangency portfolio, which maximizes the Sharpe ratio, achieves an expected return of 7.46% with volatility 9.47%, yielding a Sharpe ratio of 0.523. This significantly outperforms the equal-weight portfolio (Sharpe ratio: 0.463) and the minimum variance portfolio (Sharpe ratio: 0.214), demonstrating the value of optimization.

Table 1: Optimal Portfolio Characteristics

Portfolio	Expected Return (%)	Volatility (%)	Sharpe Ratio
Minimum Variance	3.55	4.92	0.214
Tangency (Max Sharpe)	7.46	9.47	0.523
Equal Weight	8.48	12.94	0.463
Risk-Free Asset	2.50	0.00	—

4.3 Asset Allocation Breakdown

The optimal tangency portfolio allocates capital across asset classes to maximize risk-adjusted returns. The allocation reflects each asset's expected return, volatility, and correlation with other assets in the portfolio.

Table 2: Tangency Portfolio Asset Allocation

Asset Class	Weight (%)	Expected Return (%)	Volatility (%)	Beta
Large Cap	32.1	10.5	18.0	1.61
Small Cap	0.3	12.0	25.0	1.92
International	5.9	11.2	22.0	1.75
Corp Bonds	61.7	5.5	8.0	0.61
Gov Bonds	0.0	3.2	5.0	0.26
REITs	0.0	8.5	20.0	1.25
Total	100.0	7.46	9.47	1.00

4.4 Risk Measures

Beyond mean-variance analysis, we compute downside risk measures for the tangency portfolio. Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) quantify the maximum expected loss at a given confidence level and the expected loss in extreme scenarios, respectively.

Table 3: Risk Measures for Tangency Portfolio

Risk Measure	Value
Portfolio Volatility (annual)	9.47%
Value-at-Risk (95% confidence)	8.12%
Conditional VaR (Expected Shortfall)	12.08%
Downside Deviation (vs. 2.5%)	6.70%

5 Discussion

5.1 Interpretation of Results

Example 5.1 (Portfolio Construction Insights) *The tangency portfolio demonstrates several key principles of Modern Portfolio Theory:*

- **Diversification:** *No single asset dominates; weights are distributed across multiple asset classes to reduce idiosyncratic risk.*
- **Risk-Return Trade-off:** *Higher-volatility assets like small-cap equities receive lower allocations despite higher expected returns, reflecting the quadratic penalty on variance.*
- **Correlation Structure:** *The significant allocation to bonds (low correlation with equities) enhances diversification and reduces overall portfolio variance.*
- **Sharpe Ratio Maximization:** *The tangency portfolio achieves a Sharpe ratio of 0.523, substantially exceeding the equal-weight benchmark.*

Remark 5.1 (Efficient Frontier Properties) *The efficient frontier exhibits the characteristic hyperbolic shape in mean-variance space. Portfolios below the minimum variance point are inefficient and never optimal. The capital market line (CML) connecting the risk-free asset to the tangency portfolio represents the highest attainable Sharpe ratio for any combination of risk-free and risky assets, according to the Separation Theorem.*

5.2 Beta Analysis and CAPM Implications

The computed beta coefficients reveal systematic risk characteristics. Large-cap equities exhibit $\beta \approx 1.61$, indicating slightly higher sensitivity to market movements than the overall tangency portfolio. Government bonds show $\beta \approx 0.26$, confirming their role as defensive assets with low systematic risk. According to CAPM, assets with higher betas should command higher expected returns to compensate for undiversifiable risk.

5.3 Risk Management Applications

The VaR and CVaR metrics provide practical risk management guidance. At 95% confidence, the tangency portfolio's maximum expected annual loss is 8.12%. The CVaR of 12.08% represents the expected loss conditional on exceeding VaR, offering insight into tail risk. These measures are essential for regulatory compliance and internal risk limits.

5.4 Limitations and Extensions

Remark 5.2 (Model Assumptions) *This analysis assumes:*

- *Returns are normally distributed (justifies VaR/CVaR calculations)*
- *Expected returns and covariance matrix are known and stationary*
- *No transaction costs or taxes*
- *Unlimited short-selling or borrowing at the risk-free rate (relaxed for long-only constraint)*

Extensions could incorporate estimation error, robust optimization, transaction costs, and non-normal return distributions.

6 Conclusions

This computational analysis demonstrates the practical implementation of Modern Portfolio Theory for multi-asset portfolio optimization. Key findings include:

1. The tangency portfolio achieves a Sharpe ratio of 0.523, representing a 13.2% improvement over the equal-weight benchmark through optimal mean-variance allocation.
2. The efficient frontier reveals that diversification reduces portfolio volatility to 9.47%, substantially below the weighted-average individual asset volatility of 12.08%, confirming the quantitative benefit of correlation-based risk reduction.
3. Beta analysis indicates that equities contribute systematic risk ($\beta > 1$) while fixed-income assets provide defensive characteristics ($\beta < 1$), consistent with CAPM predictions and enabling targeted risk exposure management.
4. Risk measures quantify downside exposure: $\text{VaR}_{95\%} = 8.12\%$ and $\text{CVaR} = 12.08\%$, providing actionable metrics for risk budgeting and regulatory compliance.

5. The minimum variance portfolio (4.92% volatility) prioritizes capital preservation but accepts lower expected return (3.55%) and Sharpe ratio (0.214), illustrating the fundamental risk-return trade-off.

These results validate the Markowitz framework as a foundational tool for quantitative portfolio management, demonstrating that systematic optimization delivers measurable improvements in risk-adjusted performance over naive diversification strategies.

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